AQA A-Level Further Mathematics (7367) — Topic 1.7

1.7: Polar Coordinates

1. Definition and Conversion

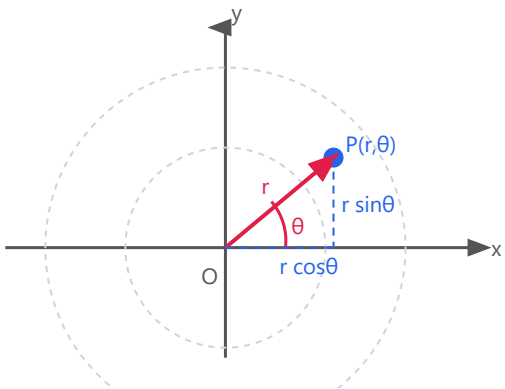
Polar coordinates: A point P is described by (r, θ) where:

- $r \geq 0$ is the distance from the origin (pole)
- θ is the angle measured anticlockwise from the initial line (positive x-axis)

Conversion formulae:

$$x = r\cos\theta, \quad y = r\sin\theta$$

$$r^2 = x^2 + y^2, \quad \tan\theta = \frac{y}{x}$$

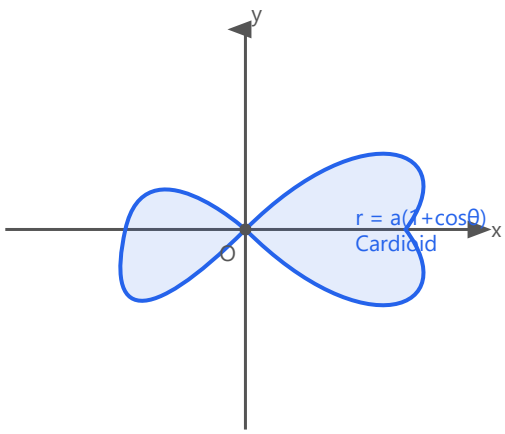


2. Common Polar Curves

Equation	Curve	Notes

$r = a$	Circle, centre O, radius a	Cartesian: $x^2 + y^2 = a^2$
$r = a\theta$	Archimedean spiral	Spacing proportional to θ
$r = a(1 + \cos\theta)$	Cardioid	Heart-shaped; passes through origin at $\theta = \pi$
$r = a(1 + \cos\theta)$	Limaçon	Dimpled or looped depending on parameter
$r = a\cos(n\theta)$	Rose curve	n petals if n odd; $2n$ petals if n even
$r^2 = a^2\cos(2\theta)$	Lemniscate of Bernoulli	Figure-8 shape; r real only when $\cos 2\theta \geq 0$
$r = a\sec\theta$	Vertical line $x = a$	Since $r\cos\theta = a$

Cardioid — SVG Illustration



3. Sketching Polar Curves

Method:

1. Find where $r = 0$ (curve passes through pole)
2. Find maximum and minimum values of r
3. Check symmetry: curve is symmetric about initial line if $r(-\theta) = r(\theta)$; about $\theta = \pi/2$ if $r(\pi - \theta) = r(\theta)$
4. Build a table of values for key angles: $0, \pi/6, \pi/4, \pi/3, \pi/2, \pi, \ldots$

5. Sketch, noting the direction of tracing as θ increases

AQA Exam Tip: AQA mark schemes for sketching award marks for: shape (1), correct intercepts on initial line (1), correct behaviour at pole/origin (1), and any labelled maximum values. Always label the maximum value of r on your sketch.

4. Area in Polar Coordinates

Area swept out by the radius vector:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} r^2 \, d\theta$$

(the area of the sector from angle α to β).

Area between two polar curves:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} (r_1^2 - r_2^2) \, d\theta$$

where $r_1 > r_2$ in $[\alpha, \beta]$.

Example: Find the total area enclosed by the cardioid $r = a(1 + \cos\theta)$.

The cardioid is traced out for $\theta \in [0, 2\pi]$, but using symmetry about the initial line:
$A = 2 \times \frac{1}{2}\int_0^{\pi} a^2(1+\cos\theta)^2 \, d\theta = a^2 \int_0^{\pi}(1 + 2\cos\theta + \cos^2\theta) \, d\theta$

Using $\cos^2\theta = \dfrac{1+\cos 2\theta}{2}$:
$= a^2 \int_0^{\pi} \left(\frac{3}{2} + 2\cos\theta + \frac{\cos 2\theta}{2}\right) d\theta$
$= a^2 \left[\frac{3\theta}{2} + 2\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{\pi} = a^2\left(\frac{3\pi}{2}\right) = \frac{3\pi a^2}{2}$

Example: Find the area of one loop of the rose curve $r = 2\cos(2\theta)$.

One petal exists where $r \geq 0$, e.g. $-\pi/4 \leq \theta \leq \pi/4$:
$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} 4\cos^2(2\theta)\,d\theta = 2\int_0^{\pi/4}(1+\cos 4\theta)\,d\theta$
$= 2\left[\theta + \frac{\sin 4\theta}{4}\right]_0^{\pi/4} = 2\left(\frac{\pi}{4}+0\right) = \frac{\pi}{2}$

AQA Exam Tip: Always identify the correct limits carefully. For a cardioid or lima\c{c}on, the full curve is $[0, 2\pi]$. For one loop of a rose, find where $r = 0$ to determine limits. Write down your limits explicitly — AQA marks this.

5. Tangents to Polar Curves

To find $\frac{dy}{dx}$ for a polar curve:

$$\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} \quad \text{where } x = r\cos\theta,\ y = r\sin\theta$$

$$\frac{dy}{d\theta} = \frac{dr}{d\theta}\sin\theta + r\cos\theta, \qquad \frac{dx}{d\theta} = \frac{dr}{d\theta}\cos\theta - r\sin\theta$$

Example: Find the tangent at the pole for $r = a(1+\cos\theta)$.

At the pole, $r = 0 \Rightarrow 1 + \cos\theta = 0 \Rightarrow \theta = \pi$.

$\frac{dr}{d\theta} = -a\sin\theta$
At $\theta = \pi$: $\frac{dy}{d\theta} = (-a\sin\pi)(\sin\pi) + 0\cdot\cos\pi = 0$
$\frac{dx}{d\theta} = (-a\sin\pi)(\cos\pi) - 0 = 0$

Indeterminate — use L'Hopital or note the tangent direction is $\theta = \pi$, i.e. horizontal.

6. Summary of Key Results

Result	Formula
Cartesian conversion	$x = r\cos\theta$, $y = r\sin\theta$
Polar conversion	$r = \sqrt{x^2 + y^2}$, $\tan\theta = y/x$
Area (polar)	$A = \frac{1}{2}\int r^2\, d\theta$
dy/dx in polar	$\dfrac{r'\sin\theta + r\cos\theta}{r'\cos\theta - r\sin\theta}$
Cardioid area	$\frac{3\pi a^2}{2}$ for $r = a(1 + \cos\theta)$

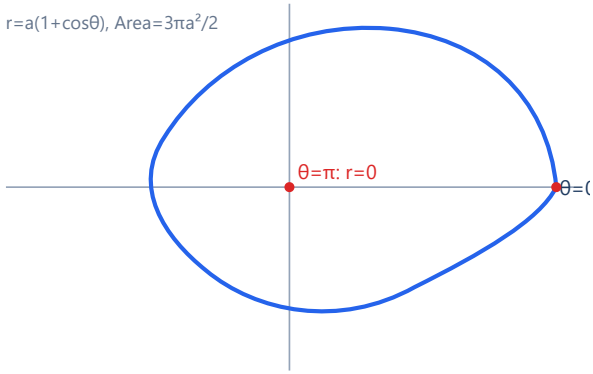
7. Practice Checklist

- Can you convert between Cartesian and polar coordinates?
- Can you sketch standard polar curves (circle, cardioid, rose, lemniscate)?
- Can you find the area enclosed by a polar curve?
- Can you find the area between two polar curves?
- Can you find the gradient dy/dx at a given point on a polar curve?

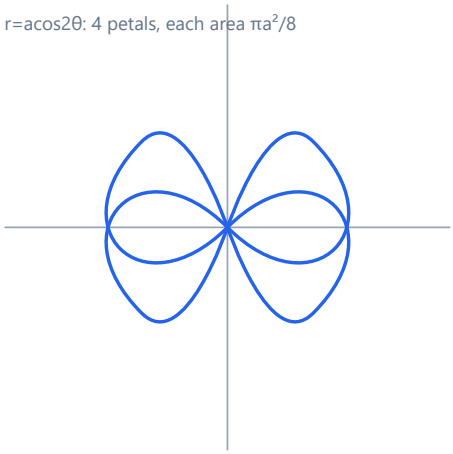
Common Polar Curves Gallery

The four most important curve families for AQA FM:

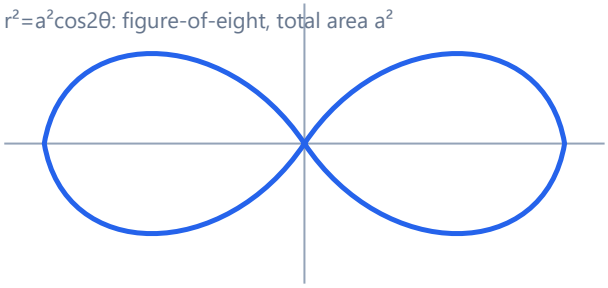
1. Cardioid: $r = a(1 + \cos\theta)$



2. Four-petal rose: $r = a\cos 2\theta$

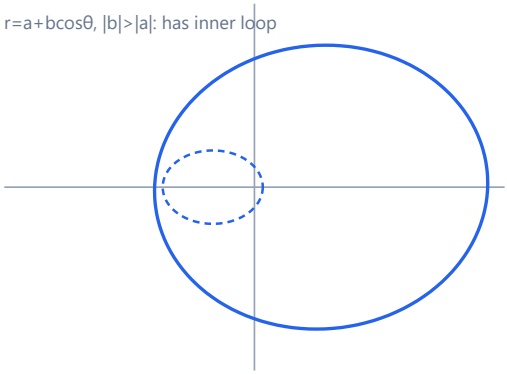


3. Lemniscate: $r^2 = a^2\cos 2\theta$



4. Limaçon: $r = a + b\cos\theta$

When $|a| > |b|$: convex limaçon (no inner loop). When $|a| < |b|$: inner loop. When $a = b$: cardioid (special case).



Tangents at the Pole

The curve $r = f(\theta)$ passes through the pole when $f(\theta_0) = 0$. The tangent at the pole has direction $\theta = \theta_0$ (i.e.\ slope $\tan\theta_0$).

Example: Find the tangent lines at the pole for $r = \cos 2\theta$. At the pole: $\cos 2\theta = 0 \Rightarrow \theta = \pi/4, 3\pi/4, 5\pi/4, 7\pi/4$. Tangent lines: $y = x$, $y = -x$, $y = x$, $y = -x$ (i.e.\ two distinct tangents $y = x$ and $y = -x$).

Area Between Two Polar Curves

The area between $r = f(\theta)$ (outer) and $r = g(\theta)$ (inner) for $\alpha \leq \theta \leq \beta$ is:

$$A = \frac{1}{2}\int_{\alpha}^{\beta} \left[f(\theta)^2 - g(\theta)^2\right] d\theta$$

Example: Find the area inside $r = 2$ but outside $r = 1 + \cos\theta$. Intersections: $1 + \cos\theta = 2 \Rightarrow \cos\theta = 1 \Rightarrow \theta = 0$. Also $r = 1 + \cos\theta = 0$ at $\theta = \pi$. For $\theta \in (0, \pi)$, the outer curve is $r = 2$ for $\theta \in (0, \pi/2)$ and inner is $r = 1 + \cos\theta$... In fact: $1 + \cos\theta \leq 2$ always since $\cos\theta \leq 1$, so $r = 2$ is always outer (except touching at $\theta = 0$).

$$A = \frac{1}{2}\int_0^{2\pi} (4 - (1 + \cos\theta)^2) d\theta = \frac{1}{2}\int_0^{2\pi} (3 - 2\cos\theta - \cos^2\theta) d\theta$$

$$= \frac{1}{2}\left[3(2\pi) - 0 - \pi\right] = \frac{1}{2}(5\pi) = \frac{5\pi}{2}$$

AQA Exam Technique — Polar Coordinates

Key tips:

- **Sketching:** Always calculate r at $\theta = 0, \pi/2, \pi, 3\pi/2$ and mark these on the diagram. Note when $r = 0$ (pole).
- **Area formula:** $A = \frac{1}{2}\int r^2\, d\theta$. The limits must cover one complete loop. For a rose $r = \cos n\theta$, one petal spans π/n radians.
- **Gradient:** Use $\dfrac{dy}{dx} = \dfrac{dy/d\theta}{dx/d\theta}$ where $x = r\cos\theta$, $y = r\sin\theta$. Differentiate fully.
- **Conversion:** To convert $r = f(\theta)$ to Cartesian, multiply both sides by r (to replace $r\cos\theta = x$, $r\sin\theta = y$, $r^2 = x^2 + y^2$).
- **Between curves:** Find intersections carefully — also check whether the origin (pole) is a common point.

Polar to Cartesian Conversions — Worked Examples

Conversion formulae: $x = r\cos\theta,\ y = r\sin\theta,\ r^2 = x^2 + y^2,\ \tan\theta = y/x$.

Example 1: Convert $r = 3\cos\theta$ to Cartesian.
Multiply both sides by r: $r^2 = 3r\cos\theta \Rightarrow x^2 + y^2 = 3x \Rightarrow \left(x - \frac{3}{2}\right)^2 + y^2 = \frac{9}{4}$.
This is a circle with centre $\left(\frac{3}{2}, 0\right)$ and radius $\frac{3}{2}$.

Example 2: Convert $r = \dfrac{2}{1 - \cos\theta}$ to Cartesian.
$r(1 - \cos\theta) = 2 \Rightarrow r - r\cos\theta = 2 \Rightarrow \sqrt{x^2 + y^2} - x = 2$.
$\sqrt{x^2 + y^2} = x + 2 \Rightarrow x^2 + y^2 = (x + 2)^2 = x^2 + 4x + 4 \Rightarrow y^2 = 4x + 4$.

Example 3: Convert Cartesian $x^2 + y^2 = 4$ to polar.
$r^2 = 4 \Rightarrow r = 2$ (circle of radius 2 centred at origin).

Polar Curves — Further Examples

Spiral of Archimedes: $r = a\theta$ — radius increases linearly with angle.
Reciprocal spiral: $r = a/\theta$ — radius decreases as angle increases.
Conic sections in polar: $r = \dfrac{l}{1 + e\cos\theta}$ where l is semi-latus rectum, e eccentricity. Circle: $e = 0$; ellipse: \(01\).

Example — Sketching $r = 1 + 2\cos\theta$ (limaçon with inner loop):
At $\theta = 0$: $r = 3$. At $\theta = \pi/2$: $r = 1$. At $\theta = 2\pi/3$: $r = 0$ (pole).
For $2\pi/3 < \theta < 4\pi/3$: $r < 0$ — this traces an inner loop.
Key feature: inner loop occurs when $r = 0$, i.e. $\cos\theta = -1/2$.

Area in Polar Coordinates — Full Examples

Example 1 — Area of one loop of $r = \cos 2\theta$:
One loop: $-\pi/4 \le \theta \le \pi/4$ (where $\cos 2\theta \ge 0$).

$$A = \frac{1}{2}\int_{-\pi/4}^{\pi/4} \cos^2 2\theta \, d\theta = \frac{1}{2}\int_{-\pi/4}^{\pi/4} \frac{1 + \cos 4\theta}{2} \, d\theta$$

$$= \frac{1}{4}\left[\theta + \frac{\sin 4\theta}{4}\right]_{-\pi/4}^{\pi/4} = \frac{1}{4} \cdot \frac{\pi}{2} = \frac{\pi}{8}$$

Example 2 — Area enclosed by $r = 2 + 2\cos\theta$ (cardioid):
Full loop: θ from 0 to 2π.

$$A = \frac{1}{2}\int_0^{2\pi} (2 + 2\cos\theta)^2 \, d\theta = \frac{1}{2}\int_0^{2\pi} 4(1 + \cos\theta)^2 \, d\theta$$

$$= 2\int_0^{2\pi} (1 + 2\cos\theta + \cos^2\theta) \, d\theta = 2\int_0^{2\pi} \left(\tfrac{3}{2} + 2\cos\theta + \tfrac{\cos 2\theta}{2}\right) d\theta$$

$$= 2\left[\frac{3\theta}{2} + 2\sin\theta + \frac{\sin 2\theta}{4}\right]_0^{2\pi} = 2 \cdot 3\pi = 6\pi$$

Example 3 — Area between two curves $r = 1$ and $r = 1 + \cos\theta$:
They intersect where $1 = 1 + \cos\theta \Rightarrow \cos\theta = 0 \Rightarrow \theta = \pm\pi/2$.
Area between curves (upper half, doubled): $A = 2 \cdot \frac{1}{2}\int_0^{\pi/2} \left[(1+\cos\theta)^2 - 1\right] d\theta$
$= \int_0^{\pi/2} (2\cos\theta + \cos^2\theta)\, d\theta = \left[2\sin\theta + \frac{\theta}{2} + \frac{\sin 2\theta}{4}\right]_0^{\pi/2} = 2 + \frac{\pi}{4}$

Tangents to Polar Curves — Full Examples

$x = r\cos\theta,\ y = r\sin\theta.$
$\frac{dy}{d\theta} = \frac{dr}{d\theta}\sin\theta + r\cos\theta, \qquad \frac{dx}{d\theta} = \frac{dr}{d\theta}\cos\theta - r\sin\theta$
Gradient: $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$. Horizontal tangent: $dy/d\theta = 0$. Vertical tangent: $dx/d\theta = 0$.

Example — Tangents to $r = 1 + \cos\theta$ at the pole:
At the pole, $r = 0 \Rightarrow 1 + \cos\theta = 0 \Rightarrow \theta = \pi$.
Tangent at pole is the line $\theta = \pi$, i.e. the negative x-axis.

Example — Horizontal tangent to $r = 1 + \cos\theta$:
$\frac{dy}{d\theta} = -\sin\theta \cdot \sin\theta + (1+\cos\theta)\cos\theta = \cos\theta + \cos^2\theta - \sin^2\theta = 2\cos^2\theta + \cos\theta - 1.$
Set to zero: $(2\cos\theta - 1)(\cos\theta + 1) = 0 \Rightarrow \cos\theta = \frac{1}{2}$ or $\cos\theta = -1$.
At $\theta = \pi/3$: $r = \frac{3}{2}$, point $\left(\frac{3}{4}, \frac{3\sqrt{3}}{4}\right)$. Horizontal tangent here.

Symmetry of Polar Curves

Symmetry	Condition
About initial line (x-axis)	$r(-\theta) = r(\theta)$, i.e. r is even function of θ

About $\theta = \pi/2$ (y-axis)	$r(\pi - \theta) = r(\theta)$
About pole (origin)	$r(\pi + \theta) = r(\theta)$ or $r(-\theta) = -r(\theta)$

Example: $r = \cos 2\theta$ is symmetric about both axes (since $\cos(-2\theta) = \cos 2\theta$ and $\cos(2(\pi - \theta)) = \cos(2\pi - 2\theta) = \cos 2\theta$). So it suffices to sketch one quadrant and reflect.

Standard Polar Curves Summary

Name	Equation	Features
Circle	$r = a$	Centre origin, radius a
Circle	$r = 2a\cos\theta$	Centre $(a, 0)$, radius a
Cardioid	$r = a(1 + \cos\theta)$	Heart-shape, passes through pole
Rose n even	$r = a\cos n\theta$	$2n$ petals
Rose n odd	$r = a\cos n\theta$	n petals
Limaçon	$r = a + b\cos\theta$	Inner loop if $b > a$
Lemniscate	$r^2 = a^2\cos 2\theta$	Figure-of-eight; exists only for $\cos 2\theta \geq 0$

Arc Length in Polar Coordinates

Arc length of polar curve $r = f(\theta)$ from α to β:

$$L = \int_{\alpha}^{\beta} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

Example — Arc length of cardioid $r = a(1 + \cos\theta)$:
$\frac{dr}{d\theta} = -a\sin\theta$.
$r^2 + \left(\frac{dr}{d\theta}\right)^2 = a^2(1+\cos\theta)^2 + a^2\sin^2\theta = a^2(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta) = a^2(2 + 2\cos\theta)$.
Using $2 + 2\cos\theta = 4\cos^2(\theta/2)$:
$L = \int_0^{2\pi} a \cdot 2|\cos(\theta/2)|\, d\theta = 4a \int_0^{\pi} \cos u\, du \cdot 2 = 8a$.

Finding Areas Using Symmetry

Example — Area of lemniscate $r^2 = 4\cos 2\theta$:
Exists when $\cos 2\theta \geq 0$, i.e. $-\pi/4 \leq \theta \leq \pi/4$ and $3\pi/4 \leq \theta \leq 5\pi/4$.
By symmetry, total area = 4 × (area for $0 \leq \theta \leq \pi/4$):
$A = 4 \cdot \frac{1}{2} \int_0^{\pi/4} 4\cos 2\theta\, d\theta = 8\left[\frac{\sin 2\theta}{2}\right]_0^{\pi/4} = 8 \cdot \frac{1}{2} = 4$

Example — Area inside $r = 3\cos\theta$ but outside $r = 1 + \cos\theta$:
Find intersections: $3\cos\theta = 1 + \cos\theta \Rightarrow 2\cos\theta = 1 \Rightarrow \theta = \pm\pi/3$.
$A = \frac{1}{2} \int_{-\pi/3}^{\pi/3} \left[(3\cos\theta)^2 - (1+\cos\theta)^2\right] d\theta$
$= \frac{1}{2} \int_{-\pi/3}^{\pi/3} (9\cos^2\theta - 1 - 2\cos\theta - \cos^2\theta)\, d\theta$
$= \frac{1}{2} \int_{-\pi/3}^{\pi/3} (8\cos^2\theta - 2\cos\theta - 1)\, d\theta$

$$= \frac{1}{2}\int_{-\pi/3}^{\pi/3}(4 + 4\cos 2\theta - 2\cos\theta - 1)\,d\theta$$
$$= \frac{1}{2}\left[3\theta + 2\sin 2\theta - 2\sin\theta\right]_{-\pi/3}^{\pi/3} = \frac{1}{2}\left(2\pi + 2\sqrt{3} - 2\sqrt{3}\right) = \pi$$

Polar Curves — Detailed Sketching Technique

Step-by-step method for sketching $r = f(\theta)$:

1. Identify the type (cardioid, rose, limaçon, circle, etc.) from the form.
2. Find when $r = 0$: these are the tangent lines at the pole.
3. Find the maximum value of r and the angles where it occurs.
4. Make a table of r at key angles: $0, \pi/6, \pi/4, \pi/3, \pi/2, \pi, 3\pi/2$.
5. Check symmetry to halve the work.
6. Note: if $r < 0$ is obtained, the curve is in the opposite direction.

Sketching $r = 2\sin 3\theta$ (three-petal rose):
$r = 0$ when $\sin 3\theta = 0 \Rightarrow \theta = 0, \pi/3, 2\pi/3, \pi$.
Maximum $r = 2$ when $\sin 3\theta = 1 \Rightarrow \theta = \pi/6, \pi/2, 5\pi/6$.
Three petals, at angles $\pi/6, \pi/2, 5\pi/6$ from the initial line. Symmetric about $\theta = \pi/2$.

Equations of Lines in Polar Form

- Half-line at angle α: $\theta = \alpha$
- Line perpendicular to initial line at distance a: $r\cos\theta = a$ (i.e. $x = a$)
- Line parallel to initial line at height b: $r\sin\theta = b$ (i.e. $y = b$)
- General line: $r(a\cos\theta + b\sin\theta) = c$

Complex Numbers and Polar Form — Connection

The polar form of a complex number $z = x + iy$ is $z = re^{i\theta} = r(\cos\theta + i\sin\theta)$, directly using polar coordinates (r, θ).
Multiplying two complex numbers in polar form: $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1+\theta_2)}$ — multiply moduli, add arguments. This is the geometric origin of the polar area and rotation connection.

Integration Applications in Polar Coordinates — Summary

Quantity	Formula
Area enclosed by curve	$\frac{1}{2}\int_\alpha^\beta r^2\,d\theta$
Area between two curves	$\frac{1}{2}\int_\alpha^\beta (r_{\text{outer}}^2 - r_{\text{inner}}^2)\,d\theta$
Arc length	$\int_\alpha^\beta \sqrt{r^2 + (r')^2}\,d\theta$
x-coordinate of centroid	$\bar{x} = \dfrac{\frac{1}{3}\int r^3\cos\theta\,d\theta}{\frac{1}{2}\int r^2\,d\theta}$

Worked Example — Tangent Lines at the Pole

At the pole ($r = 0$), the tangent direction is $\theta = \theta_0$ where $r(\theta_0) = 0$. If the curve passes through the pole at angle θ_0, the tangent line is $\theta = \theta_0$.

Example — $r = \sin 2\theta$:
$r = 0$ when $\sin 2\theta = 0 \Rightarrow \theta = 0, \pi/2, \pi, 3\pi/2$.
Tangent lines at the pole: $\theta = 0$ (the initial line), $\theta = \pi/2$, $\theta = \pi$, $\theta = 3\pi/2$.
These are the x and y axes — the four-petal rose is tangent to the coordinate axes at the origin.

Polar Gradient — Worked Calculation

Find the gradient of $r = 1 + \cos\theta$ at $\theta = \pi/2$:

$x = r\cos\theta = (1+\cos\theta)\cos\theta = \cos\theta + \cos^2\theta.$

$y = r\sin\theta = (1+\cos\theta)\sin\theta = \sin\theta + \sin\theta\cos\theta.$

$\dfrac{dx}{d\theta} = -\sin\theta - 2\sin\theta\cos\theta = -\sin\theta(1+2\cos\theta).$

$\dfrac{dy}{d\theta} = \cos\theta + \cos^2\theta - \sin^2\theta = \cos\theta + \cos 2\theta.$

At $\theta = \pi/2$: $\dfrac{dy}{d\theta} = 0 + \cos\pi = -1.$ $\dfrac{dx}{d\theta} = -1\cdot(1+0) = -1.$

Gradient $= \dfrac{-1}{-1} = 1.$

Polar Coordinates — AQA Mark Scheme Examples

Typical 6-mark area question structure:

"Find the exact area enclosed by $r = 3\sin 2\theta$ for the petal in the first quadrant."

- [B1] Correct limits: $0 \le \theta \le \pi/2$ (petal from $\theta = 0$ to $\theta = \pi/2$)
- [M1] Write area as $\dfrac{1}{2}\displaystyle\int_0^{\pi/2} 9\sin^2 2\theta\, d\theta$
- [M1] Use double angle formula: $\sin^2 2\theta = \dfrac{1-\cos 4\theta}{2}$
- [A1] Integrate: $\dfrac{9}{2}\left[\dfrac{\theta}{2} - \dfrac{\sin 4\theta}{8}\right]_0^{\pi/2}$
- [A1] Apply limits: $\dfrac{9}{2}\cdot\dfrac{\pi}{4} = \dfrac{9\pi}{8}$

Converting Between Polar and Parametric Forms

Polar curve $r = f(\theta)$ can be expressed in parametric form as:

$$x(\theta) = f(\theta)\cos\theta, \qquad y(\theta) = f(\theta)\sin\theta$$

This allows standard calculus techniques (area, arc length) to be applied directly.

Example: Express $r = 2\cos\theta$ in parametric form and verify it is a circle.
$x = 2\cos^2\theta,\ y = 2\cos\theta\sin\theta = \sin 2\theta$.
Alternatively, from Cartesian form: $r = 2\cos\theta \Rightarrow x^2 + y^2 = 2x \Rightarrow (x-1)^2 + y^2 = 1$. Circle with centre $(1, 0)$, radius 1. ✓

Polar Coordinates — Common Errors

- **Wrong limits:** For one petal of $r = \cos n\theta$, use the range where $r \geq 0$ — from where $r = 0$ to the next zero. Do NOT use 0 to 2π for a single loop.
- **Missing factor of $\frac{1}{2}$:** The area formula is $\frac{1}{2}\int r^2\,d\theta$, not $\int r^2\,d\theta$.
- **Negative r:** When $r < 0$, the point is in the opposite direction. This traces a second copy — include it only if it produces a distinct curve.
- **Cartesian conversion:** Multiply $r = f(\theta)$ through by r before substituting $r^2 = x^2 + y^2$ and $r\cos\theta = x$.
- **Intersections:** Polar intersections may not be visible from solving $r_1 = r_2$ — the pole may be a common point reached at different θ values.

Polar Coordinates — Further Worked Examples

Example — Show that $r = 2a\sin\theta$ is a circle and find its radius:
Multiply by r: $r^2 = 2ar\sin\theta \Rightarrow x^2 + y^2 = 2ay \Rightarrow x^2 + (y-a)^2 = a^2$.
Circle with centre $(0, a)$ and radius $|a|$.

Example — Find points where $r = 4\cos\theta$ and $r = 4\sin\theta$ intersect:
Setting equal: $\cos\theta = \sin\theta \Rightarrow \tan\theta = 1 \Rightarrow \theta = \pi/4$.
At $\theta = \pi/4$: $r = 4\cos(\pi/4) = 2\sqrt{2}$. Cartesian: $(\sqrt{2}\cdot 2\sqrt{2}/\sqrt{2}, \ldots) = (2, 2)$.

Also check the pole: $r = 0$ for $r = 4\cos\theta$ when $\theta = \pi/2$; for $r = 4\sin\theta$ when $\theta = 0$. So the pole is also a common point.

Example — Volume of revolution using polar form:

The curve $r = a$ (circle) rotated about initial line gives sphere. Volume:

$V = \pi \int_{-a}^{a} (a^2 - x^2)\, dx = \pi \left[a^2 x - \frac{x^3}{3}\right]_{-a}^{a} = \frac{4\pi a^3}{3}$ (standard result).

Polar Area — Key Integration Patterns

Common integrals arising in polar area calculations:

Integrand	Result
$\int \cos^2\theta\, d\theta$	$\frac{\theta}{2} + \frac{\sin 2\theta}{4} + C$
$\int \sin^2\theta\, d\theta$	$\frac{\theta}{2} - \frac{\sin 2\theta}{4} + C$
$\int \cos^2 n\theta\, d\theta$	$\frac{\theta}{2} + \frac{\sin 2n\theta}{4n} + C$
$\int (1+\cos\theta)^2\, d\theta$	$\frac{3\theta}{2} + 2\sin\theta + \frac{\sin 2\theta}{4} + C$

Quick Revision: Key Polar Results

- Cardioid $r = a(1+\cos\theta)$: total area $= \frac{3\pi a^2}{2}$, total arc length $= 8a$.
- Circle $r = 2a\cos\theta$: area $= \pi a^2$ (as expected for radius a).
- Rose $r = a\cos n\theta$: area of each petal $= \frac{\pi a^2}{4n}$; total area $= \frac{\pi a^2}{4}$ (all petals combined, regardless of n).

- Lemniscate $r^2 = a^2 \cos 2\theta$: total area $= a^2$.
- At the pole: the tangent direction $\theta = \theta_0$ where $r(\theta_0) = 0$.
- Cartesian gradient: $\dfrac{dy}{dx} = \dfrac{r' + r\cos\theta \cdot (1/\sin\theta)}{r' - r\sin\theta \cdot (1/\cos\theta)}$ — or better, compute $dy/d\theta$ and $dx/d\theta$ separately.
- When converting $r = f(\theta)$ to Cartesian, if you cannot simplify the expression, leave it in the form $r^2 = \ldots$ with $r^2 = x^2 + y^2$ substituted — this is acceptable in AQA answers.
- The formula sheet gives the area formula $\frac{1}{2}\int_\alpha^\beta r^2\, d\theta$ — always double-check your limits are in radians and that they describe exactly one complete region.
- For AQA proof questions involving polar curves: begin from $x = r\cos\theta,\ y = r\sin\theta$ and differentiate, showing all intermediate steps for full marks.